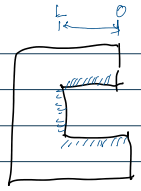
Lecture 8

Thursday, 2 February 2023 12:02 PM

$$\underset{\sim}{D}\frac{d^2\underset{\sim}{c}}{dx^2} + \frac{P}{A}\hat{K}\underset{\sim}{c} = 0 \quad ; \quad 0 < x < l$$



$$l = \underset{\sim}{D}A \left.\frac{dc}{dx}\right|_{x=0}$$

$$c_0 = \frac{P}{\hat{K}c_0}$$

$$NPl\hat{K}c_0 = \left[\underset{\sim}{D}A\frac{dc}{dx}\right]_{x=0}$$

Non-dimensionalizing : (Positive definite w/ the negative sign) Making

$$\xi = \frac{x}{l} \quad ; \quad \phi^2 = \frac{-Pl^2\underset{\sim}{D}^{-1}\hat{K}}{A}$$

$$\frac{d^2c}{d\xi^2} = \phi^2 c \Rightarrow c = a'\cos\phi\xi + b'\sinh\phi\xi = a\cosh\phi(1-\xi) + b'\sinh\phi(1-\xi) \quad ; \text{ BC: } \frac{dc}{d\xi} = 0 \text{ @ } \xi = 1$$

$$c_0 = (\cosh\phi)b_1 \text{ @ } \xi = 0$$

Sylvester's decomposition/expansion of a matrix:

$$\underset{(n\times n)}{\underset{\sim}{A}} = \sum_{j=1}^{n} \lambda_j \underset{\sim}{P_j}$$

$$\underset{\sim}{P_j} = \frac{\text{adj}(\lambda_j\underset{\sim}{I} - \underset{\sim}{A})}{\prod_{r=1,j}^{n}(\lambda_j - \lambda_r)} \rightarrow \text{projection matrix} \qquad \underset{\sim}{P_j}^2 = \underset{\sim}{P_j}$$

$(\underset{\sim}{P_i}\underset{\sim}{P_j} = 0 \quad i \neq j)$

$(\lambda_1, \lambda_2 \ldots \lambda_n)$

Let $\underset{\sim}{W_1}^T, \underset{\sim}{W_2}^T, \ldots \underset{\sim}{W_n}^T$ be the row eigenvectors of $\underset{\sim}{A}$

Let $\underset{\sim}{v_1}, \underset{\sim}{v_2}, \ldots \underset{\sim}{v_n}$ be the column eigenvectors of $\underset{\sim}{A}$

$$\underset{[ran)}{\underset{\sim}{W_j}^T}\underset{(nxn)}{\underset{\sim}{A}} = \lambda_j \underset{\sim}{W_j}^T \quad ; \quad \underset{\sim}{A}\underset{\sim}{v_j} = \lambda_j \underset{\sim}{v_j}$$

$$\underset{\sim}{P_j} = \frac{\underset{\sim}{v_j}\cdot\underset{\sim}{W_j}^T}{\underset{\sim}{W_j}^T\underset{\sim}{v_j}} \qquad \underset{\sim}{W_j}^T\underset{\sim}{v_k} = 0 \quad j \neq k$$

Biorthogonality

$$\underset{\sim}{P_j}^2 = \frac{(\underset{\sim}{v_j}\underset{\sim}{W_j}^T)(\underset{\sim}{v_j}\underset{\sim}{W_j}^T)}{(\underset{\sim}{W_j}^T\underset{\sim}{v_j})(\underset{\sim}{W_j}^T\underset{\sim}{v_j})} = \frac{\underset{\sim}{v_j}(\underset{\sim}{W_j}^T\underset{\sim}{v_j})\underset{\sim}{W_j}^T}{(\underset{\sim}{W_j}^T\underset{\sim}{v_j})(\underset{\sim}{W_j}^T\underset{\sim}{v_j})} = \underset{\sim}{P_j}$$

$$\underset{\sim}{c}^A = \sum_{j=1}^{N} e^{\lambda_j}\underset{\sim}{P_j}$$

$$\cosh\underset{\sim}{\phi} = \sum_{j=1}^{n}\cosh\lambda_j\underset{\sim}{P_j} \quad ; \quad \underset{\sim}{\phi}^{-1} = \sum\lambda_j^{-1}\underset{\sim}{P_j}$$

$$[\cosh\underset{\sim}{\phi}]^{-1} = \underset{\lambda=0}{\underset{\sim}{P_1}} + \sum_{j=2}^{n}(\cosh\lambda_j)^{-1}\underset{\sim}{P_j}$$

(since $\cosh 0 = 1$)

Temperature Effect on Apparent Activation Energy

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$\ln k$ vs. $\frac{1}{T}$; if no transport effects : straight line $\frac{E}{R}$

However, diffusion limitations distort this behaviour

$\hat{r}_{obs} = \hat{r}\eta$ (assuming uniform temp.)

$\hat{r} = Ae^{-E/RT} \cdot c_s$; c_s = reactant

@ fixed c_s: $\frac{d\ln \hat{r}}{dT^{-1}} = \frac{d\ln \hat{k}}{dT^{-1}} = \frac{E}{R}$